

SANJAY MURALIDHAR

Aspiring Quantitative & Financial Markets Analyst

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SUMMARY

Computer Science & Engineering (Data Science) undergraduate with a strong quantitative and analytical foundation, built through a self-driven project forecasting NIFTY price movement from financial news sentiment and historical price data. Comfortable with statistical modeling, pattern identification, and translating unstructured data into decision-ready signals. Eager to apply and sharpen this analytical rigor in a fast-paced derivatives trading environment.

TECHNICAL SKILLS

- Quantitative & Statistical Analysis: feature engineering, statistical model evaluation & validation, target-leakage detection, pattern identification
- Data Analysis: Pandas, NumPy, Scikit-learn, NLTK, sentiment & time-series feature extraction
- Programming: Python, C
- Applied AI/GenAI: LangChain, Groq API, LLM-based application development
- Data & Databases: MySQL, data ingestion & parsing pipelines
- Tools: Git, GitHub, Streamlit, Jupyter Notebook

EDUCATION

Manipal Institute of Technology, Bengaluru

Expected 2027

B.Tech, Computer Science & Engineering (Data Science) | CGPA: 7.25/10

Relevant Coursework: Probability & Statistics, Machine Learning, Data Structures & Algorithms, Database Management Systems

PROJECTS

Market Sentiment-Based Stock Movement Prediction System | Personal Project

2025

- Researched and built an end-to-end system (Python, Pandas, Scikit-learn, NLTK) to identify patterns linking financial news sentiment to NIFTY price movement
- Assimilated financial news and historical price data into lagged sentiment, price-difference, and rolling-window features to quantify market sentiment for decision-making
- Developed and iterated on prediction strategies, improving performance from a Linear Regression baseline ($R^2 = 0.01$) to a Gradient Boosting model ($R^2 = 0.30$)
- Validated findings through feature importance analysis and target-leakage detection to ensure signals were genuine and not statistical artifacts
- Deployed the system as a Streamlit app for real-time sentiment-driven market analysis

EXPERIENCE

Software Development Intern | Tech Mahindra

Jun 2025 – Jul 2025

- Built a data-driven evaluation platform in Python, converting unstructured text into structured, comparable compatibility scores
- Designed a document ingestion and parsing pipeline to extract structured information from unstructured resumes and job descriptions
- Integrated LLM-based analysis to generate real-time, data-backed feedback and recommendations
- Built an interactive interface for real-time evaluation, translating analytical output into a usable decision-support tool